

Yukawa potential

Find $d\sigma$ for the Yukawa potential

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r} \exp(-\epsilon r)$$

§1

Start with the Born approximation

$$f(\theta, \phi) = -\frac{m}{2\pi\hbar^2} \int \exp(i\mathbf{k} \cdot \mathbf{r}) V(\mathbf{r}) d\mathbf{r}$$

where $\mathbf{k}$ is momentum transfer

$$\mathbf{k} = \mathbf{k}_i - \mathbf{k}_f$$

and

$$k^2 = |\mathbf{k}|^2 = \frac{4mE(1 - \cos\theta)}{\hbar^2}$$

Convert to polar coordinates.

$$f(\theta, \phi) = -\frac{m}{2\pi\hbar^2} \int_0^\infty \int_0^\pi \int_0^{2\pi} \exp(ikr \cos\theta) V(r, \theta, \phi) r^2 \sin\theta dr d\theta d\phi$$

Substitute

$$V(r, \theta, \phi) = -\frac{Ze^2}{4\pi\epsilon_0 r} \exp(-\epsilon r)$$

to obtain

$$f(\theta, \phi) = \frac{mZe^2}{8\pi^2\epsilon_0\hbar^2} \int_0^\infty \int_0^\pi \int_0^{2\pi} \exp(ikr \cos\theta - \epsilon r) r \sin\theta dr d\theta d\phi$$

Integrate over ϕ (multiply by 2π).

$$f(\theta, \phi) = \frac{mZe^2}{4\pi\epsilon_0\hbar^2} \int_0^\infty \int_0^\pi \exp(ikr \cos\theta - \epsilon r) r \sin\theta dr d\theta$$

For change of variable let $y = \cos\theta$ and $dy = -\sin\theta d\theta$. The minus sign in dy is canceled by interchanging integration limits $\cos(0) = 1$ and $\cos(\pi) = -1$.

$$f(\theta, \phi) = \frac{mZe^2}{4\pi\epsilon_0\hbar^2} \int_0^\infty \int_{-1}^1 \exp(ikry - \epsilon r) r dr dy$$

Solve the integral over y .

$$f(\theta, \phi) = \frac{mZe^2}{4\pi\epsilon_0\hbar^2} \int_0^\infty \left[\frac{1}{ikr} \exp(ikry - \epsilon r) \right]_{y=-1}^{y=1} r dr$$

Cancel r and evaluate the limits.

$$f(\theta, \phi) = \frac{mZe^2}{4\pi\epsilon_0\hbar^2} \frac{1}{ik} \int_0^\infty [\exp(ikr - \epsilon r) - \exp(-ikr - \epsilon r)] dr$$

Solve the integral over r .

$$f(\theta, \phi) = \frac{mZe^2}{4\pi\epsilon_0\hbar^2} \frac{1}{ik} \left[\frac{1}{ik - \epsilon} \exp(ikr - \epsilon r) + \frac{1}{ik + \epsilon} \exp(-ikr - \epsilon r) \right]_{r=0}^{r=\infty}$$

Evaluate the limits. The exponentials vanish at the upper limit.

$$f(\theta, \phi) = \frac{mZe^2}{4\pi\epsilon_0\hbar^2} \frac{1}{ik} \left[-\frac{1}{ik - \epsilon} - \frac{1}{ik + \epsilon} \right]$$

By the identity

$$\frac{1}{ik} \left[-\frac{1}{ik - \epsilon} - \frac{1}{ik + \epsilon} \right] = \frac{2}{k^2 + \epsilon^2} \quad (1)$$

we have

$$f(\theta, \phi) = \frac{mZe^2}{2\pi\epsilon_0\hbar^2(k^2 + \epsilon^2)} \quad (2)$$

Substitute for k^2 .

$$f(\theta, \phi) = \frac{mZe^2}{2\pi\epsilon_0[4mE(1 - \cos \theta) + \epsilon^2\hbar^2]}$$

Hence

$$\frac{d\sigma}{d\Omega} = |f(\theta, \phi)|^2$$

§2

For $\epsilon = 0$ the scattering amplitude simplifies as

$$f(\theta, \phi) = \frac{Ze^2}{8\pi\epsilon_0 E(1 - \cos \theta)}$$

Substitute $e^2 = 4\pi\epsilon_0\alpha\hbar c$ to obtain

$$f(\theta, \phi) = \frac{Z\alpha\hbar c}{2E(1 - \cos \theta)}$$

which is the amplitude for Rutherford scattering.

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